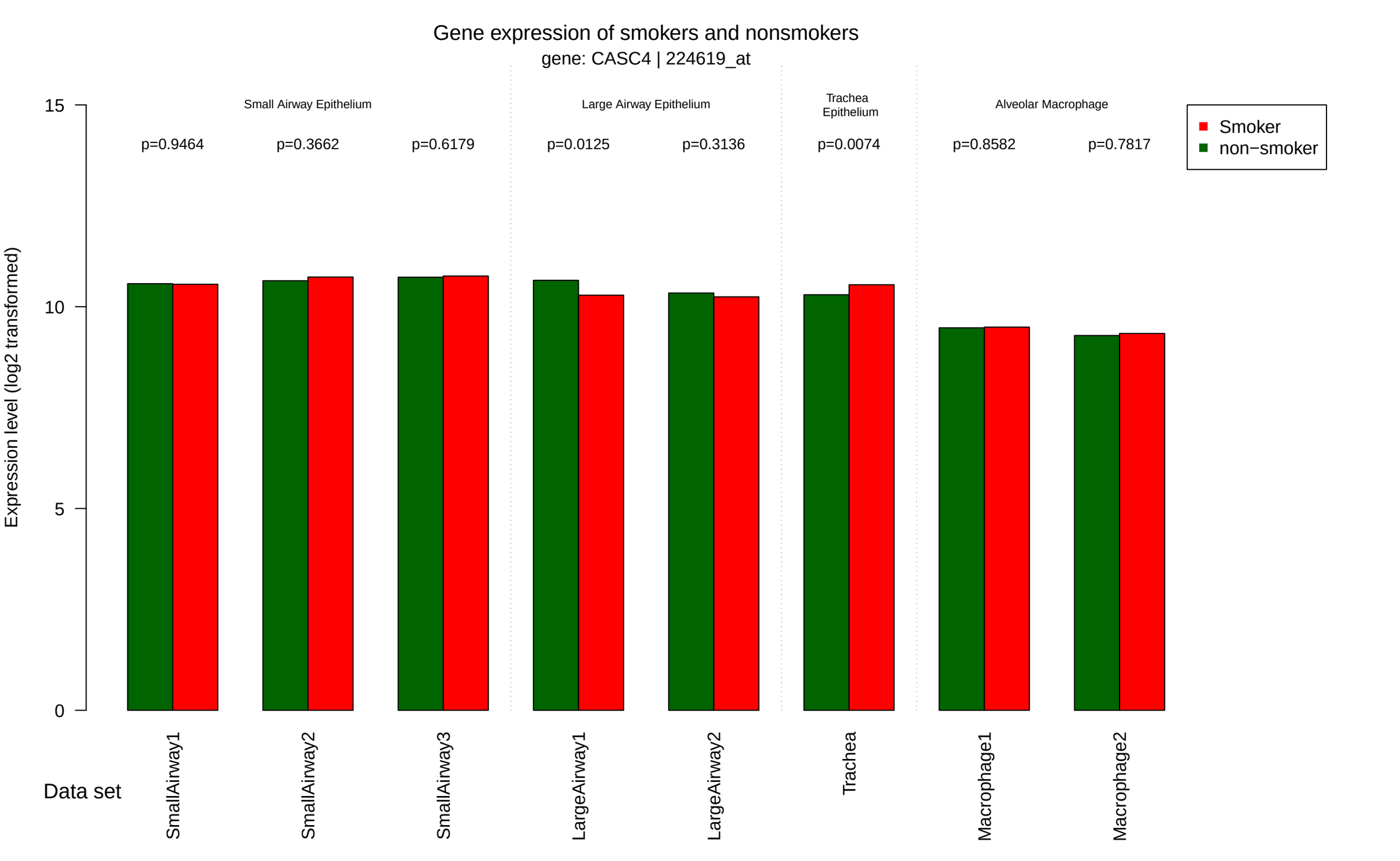
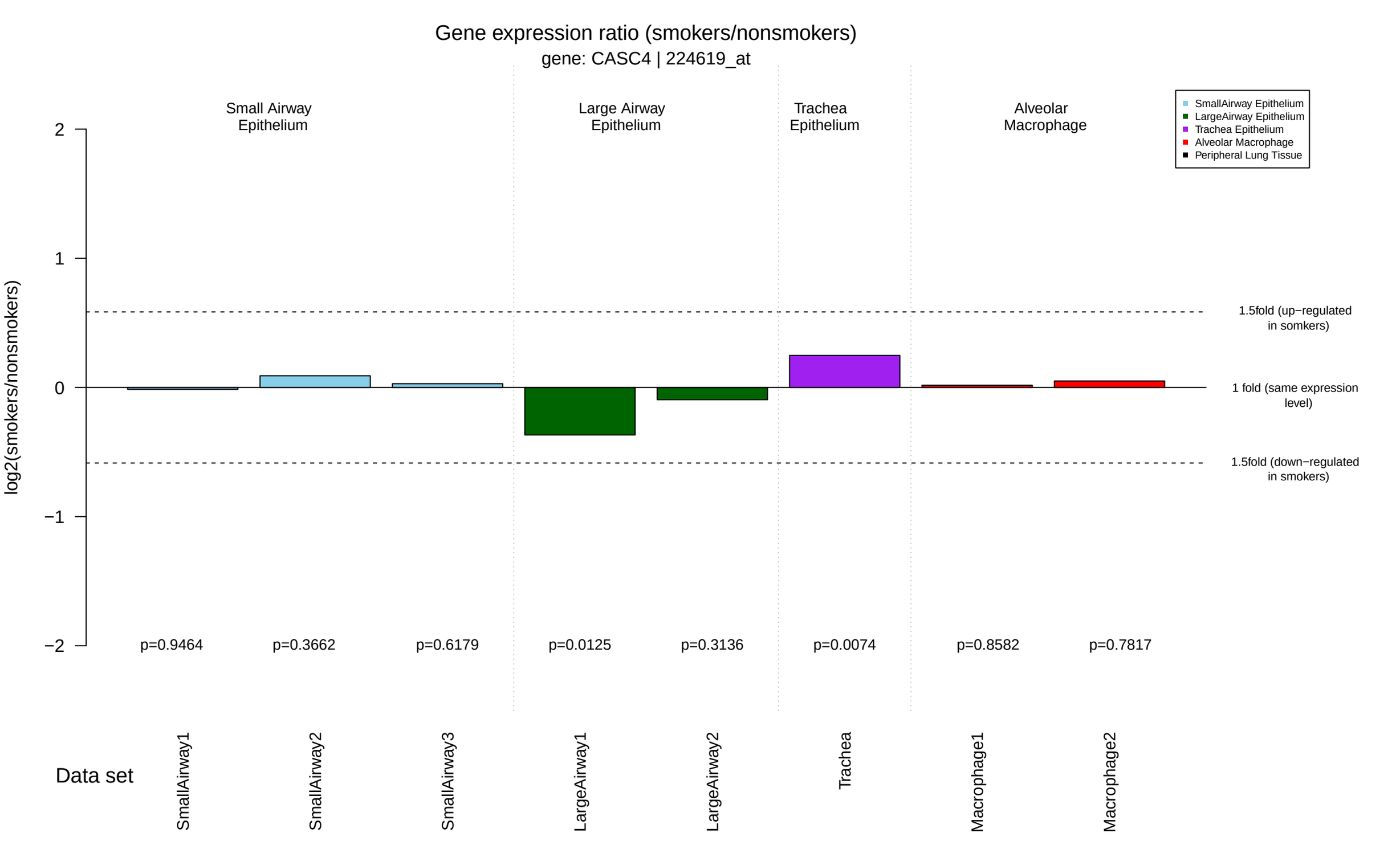
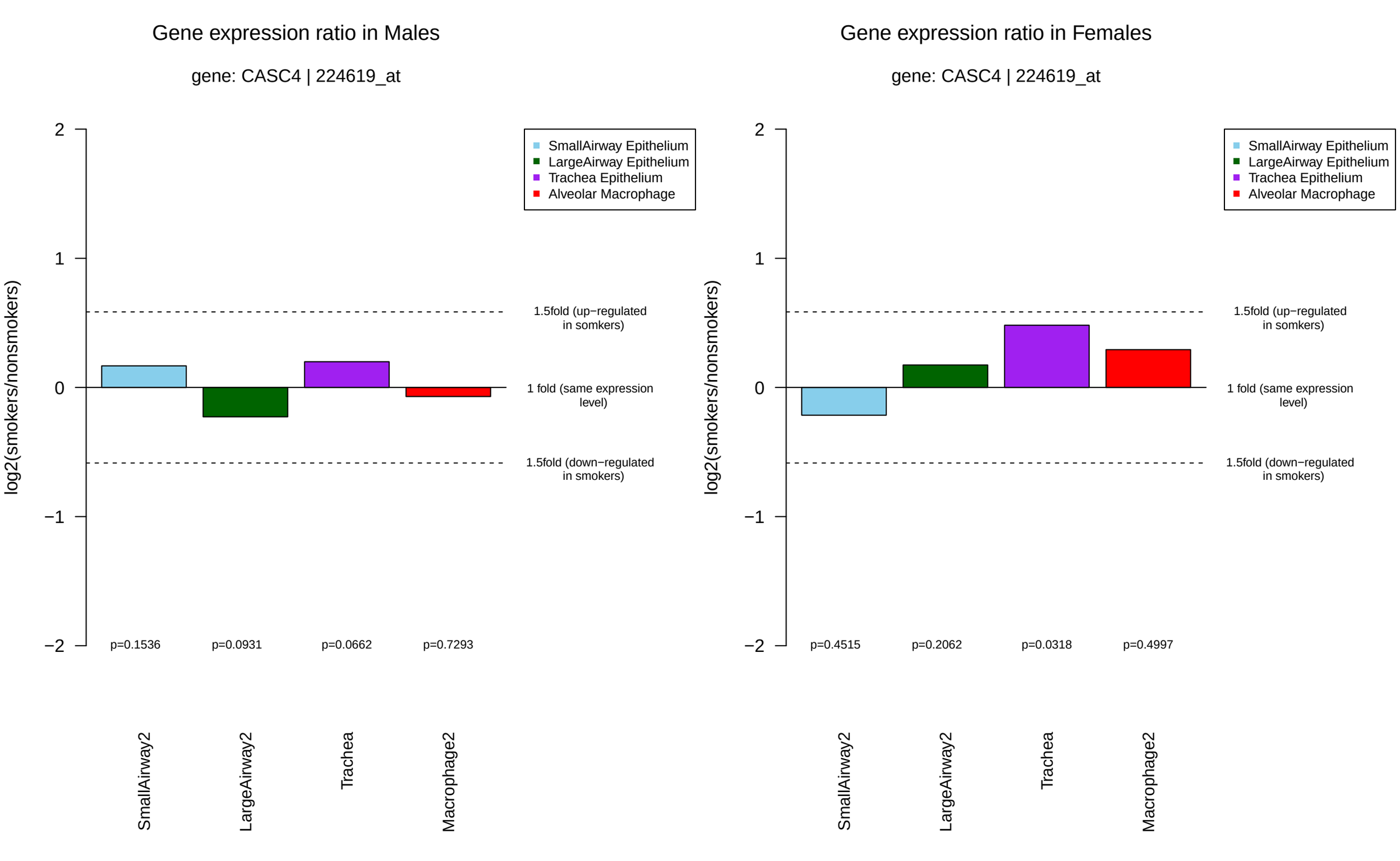
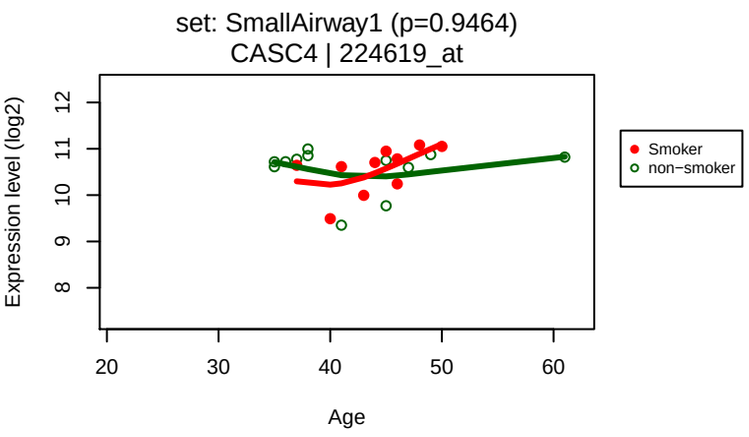
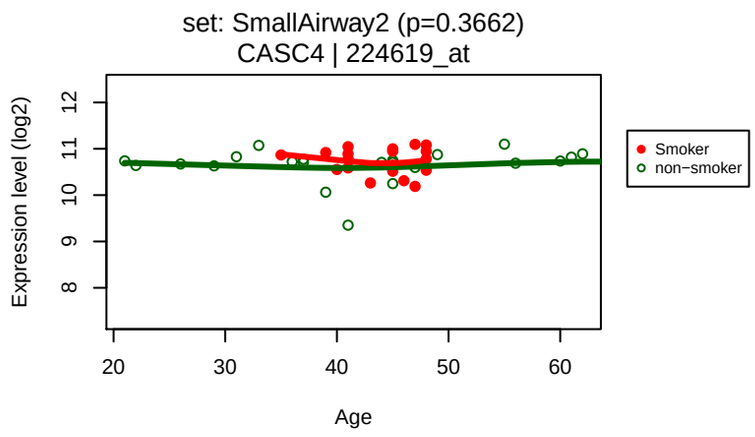
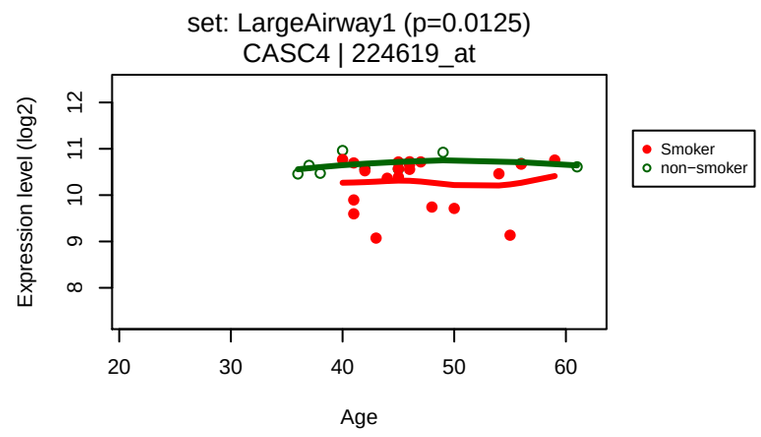

Gene expression of smokers and nonsmokers
gene: CASC4 | 224619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9464
p=0.3662
p=0.6179
p=0.0125
p=0.3136
p=0.0074
p=0.8582
p=0.7817
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

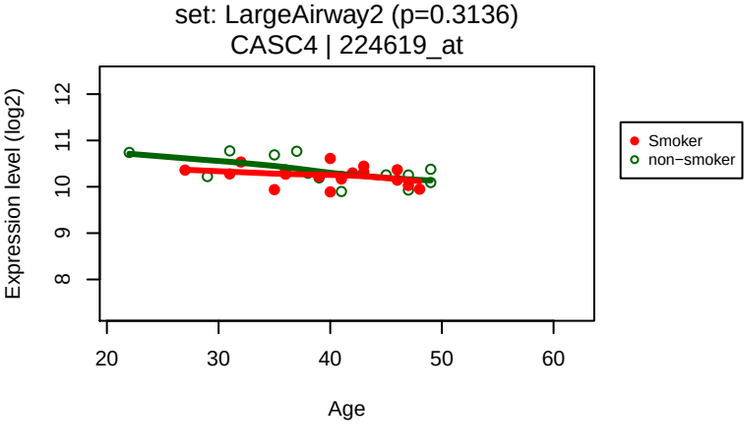

Gene expression ratio (smokers/nonsmokers)
gene: CASC4 | 224619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9464
p=0.3662
p=0.6179
p=0.0125
p=0.3136
p=0.0074
p=0.8582
p=0.7817
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

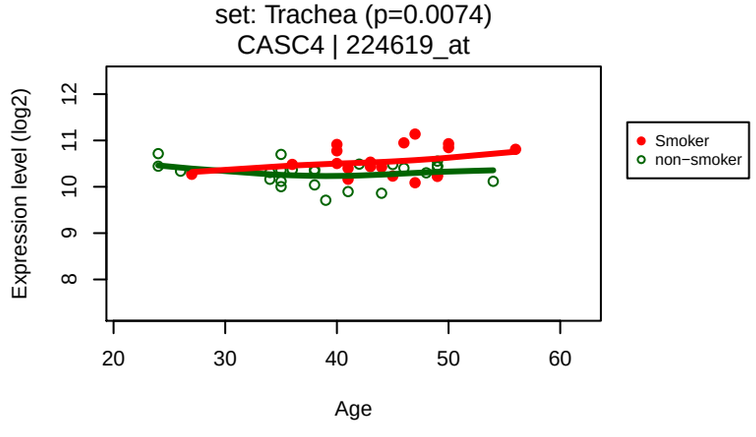

Gene expression ratio in Males
gene: CASC4 | 224619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1536
p=0.0931
p=0.0662
p=0.7293
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASC4 | 224619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4515
p=0.2062
p=0.0318
p=0.4997
SmallAirway2
LargeAirway2
Trachea
Macrophage2

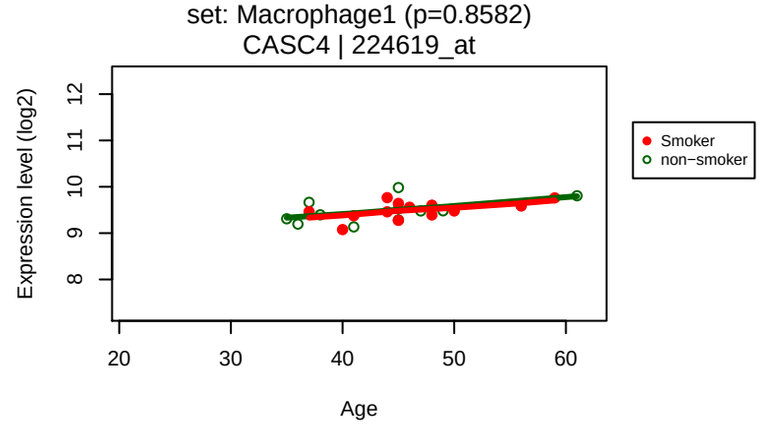

set: SmallAirway1 (p=0.9464)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker

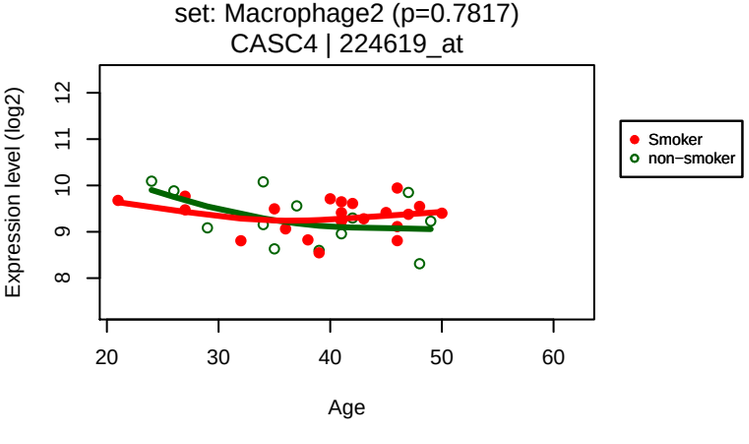

set: SmallAirway2 (p=0.3662)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker

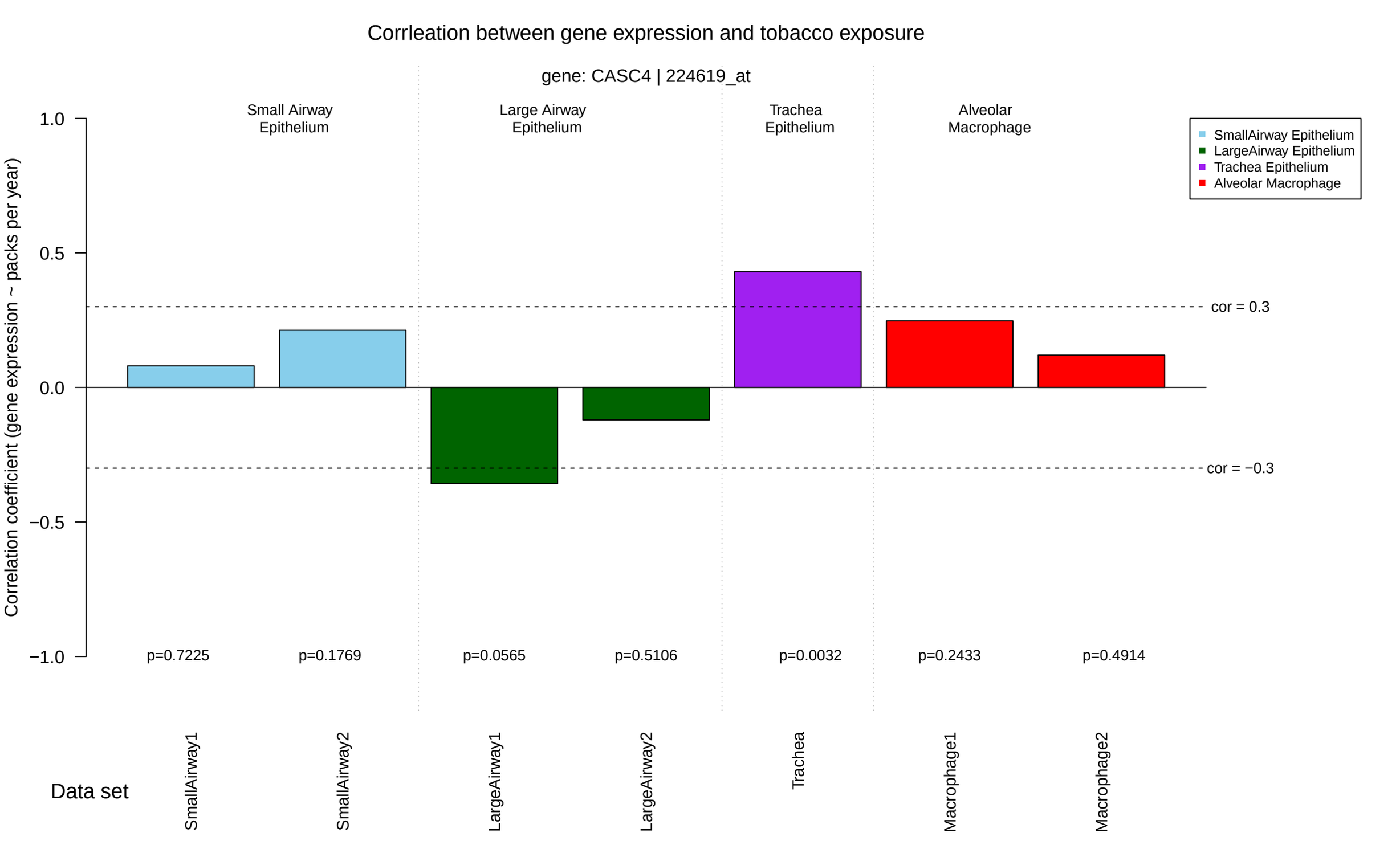

set: LargeAirway1 (p=0.0125)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3136)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0074)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8582)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7817)
CASC4 | 224619_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASC4 | 224619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7225
p=0.1769
p=0.0565
p=0.5106
p=0.0032
p=0.2433
p=0.4914
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2